

1 Summary

Model (Run-name): model3 (test)

Iteration: 6

TAC: 62852.02187721361

Took: 42.666314125061035s

Total Time: 146.5620596408844s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

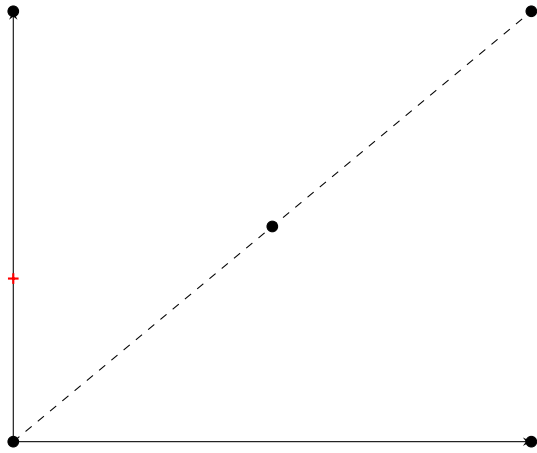
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{2,1,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{4,3,2}$	1539.72
$z_{2,2,1}$	1.0	$dt_{4,3,3}$	111.1	$q_{5,1,2}$	229.55
$z_{3,5,1}$	1.0	$dt_{5,1,2}$	38.9	$q_{5,4,2}$	1634.85
$z_{4,3,2}$	1.0	$dt_{5,1,3}$	33.56666666666667	$q_{cu,1}$	586.9599999999999
$z_{5,1,2}$	1.0	$dt_{5,4,2}$	22.2	$q_{cu,3}$	839.9899999999994
$z_{5,4,2}$	1.0	$dt_{5,4,3}$	11.366666666666667	$q_{cu,5}$	495.010000000000045
$z_{cu,1}$	1.0	$dt_{cu,1}$	120.0	$t_{1,1}^{(H)}$	160.0
$z_{cu,3}$	1.0	$dt_{cu,3}$	82.35608108108104	$t_{1,2}^{(H)}$	160.0
$z_{cu,5}$	1.0	$dt_{cu,5}$	53.56666666666667	$t_{1,3}^{(H)}$	160.0
$A_{2,1,2}^{\beta}$	0.5475650289437819	$f_{2,1,2}^H$	10.6	$t_{2,1}^{(H)}$	248.9
$A_{2,2,1}^{\beta}$	5.363812231480782	$f_{2,2,1}^H$	5.825472547254725	$t_{2,2}^{(H)}$	187.8424528301887
$A_{3,5,1}^{\beta}$	12.265236803290023	$f_{3,5,1}^H$	9.585909373060215	$t_{2,3}^{(H)}$	137.8
$A_{4,3,2}^{\beta}$	6.675846741185245	$f_{4,3,2}^H$	12.6	$t_{3,1}^{(H)}$	226.7
$A_{5,1,2}^{\beta}$	0.3222539571517902	$f_{5,1,2}^H$	1.7220555138784688	$t_{3,2}^{(H)}$	122.35608108108104
$A_{5,4,2}^{\beta}$	15.629519363344574	$f_{5,4,2}^H$	15.977944486121537	$t_{3,3}^{(H)}$	122.35608108108104
$A_{cu,1}^{\beta}$	3.357327282820013	$f_{2,1,2}^C$	5.304500000000001	$t_{4,1}^{(H)}$	271.1
$A_{cu,3}^{\beta}$	5.422975723880819	$f_{2,2,1}^C$	6.1	$t_{4,2}^{(H)}$	271.1
$A_{cu,5}^{\beta}$	2.692325040598929	$f_{3,5,1}^C$	13.9	$t_{4,3}^{(H)}$	148.9
$A_{2,1,2}$	3.303646498302869	$f_{4,3,2}^C$	8.4	$t_{5,1}^{(H)}$	198.9
$A_{2,2,1}$	16.512573095835464	$f_{5,1,2}^C$	2.2954999999999998	$t_{5,2}^{(H)}$	198.9
$A_{3,5,1}$	71.16597394333036	$f_{5,4,2}^C$	17.3	$t_{5,3}^{(H)}$	93.56666666666666
$A_{4,3,2}$	23.668855770059224	$RecLMTD_{2,1,2}$	0.023339478053072158	$t_{1,1}^{(C)}$	160.0
$A_{5,1,2}$	1.9442679879727263	$RecLMTD_{2,2,1}$	0.021686449732637236	$t_{1,2}^{(C)}$	160.0
$A_{5,4,2}$	106.27900213658211	$RecLMTD_{3,5,1}$	0.039170802020236345	$t_{1,3}^{(C)}$	60.0
$A_{cu,1}$	7.527593053896052	$RecLMTD_{4,3,2}$	0.013066354534948133	$t_{2,1}^{(C)}$	221.7
$A_{cu,3}$	16.738997089244634	$RecLMTD_{5,1,2}$	0.02032666086635175	$t_{2,2}^{(C)}$	115.6
$A_{cu,5}$	6.717165864481	$RecLMTD_{5,4,2}$	0.0552571500847753	$t_{2,3}^{(C)}$	115.6
$dt_{2,1,2}$	10.0	$RecLMTD_{cu,1}$	0.010901005342462312	$t_{3,1}^{(C)}$	221.1
$dt_{2,1,3}$	77.8	$RecLMTD_{cu,3}$	0.016938472512598907	$t_{3,2}^{(C)}$	221.1
$dt_{2,2,1}$	27.2	$RecLMTD_{cu,5}$	0.021054380844441306	$t_{3,3}^{(C)}$	37.8
$dt_{2,2,2}$	72.24245283018871	$q_{2,1,2}$	530.45	$t_{4,1}^{(C)}$	176.7
$dt_{3,5,1}$	22.3	$q_{2,2,1}$	647.2099999999999	$t_{4,2}^{(C)}$	176.7
$dt_{3,5,2}$	29.056081081081047	$q_{3,5,1}$	1544.29000000000004	$t_{4,3}^{(C)}$	82.2

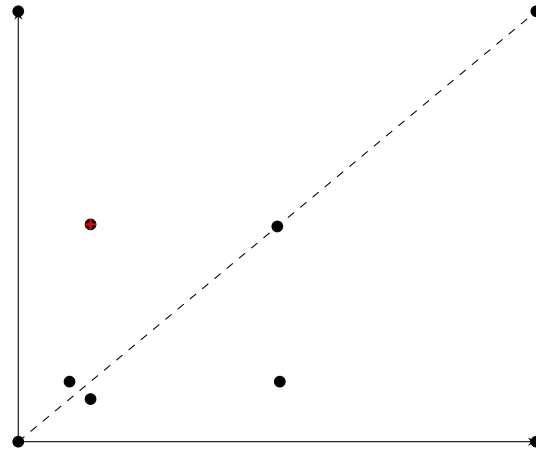
Variable	Value	Variable	Value
$t_{5,1}^{(C)}$	204.4	$b_{2,1,2}^{C,\text{out}}$	848.72
$t_{5,2}^{(C)}$	93.3	$b_{2,2,1}^{C,\text{out}}$	1352.37
$t_{5,3}^{(C)}$	93.3	$b_{3,5,1}^{C,\text{out}}$	2841.1600000000003
$t_{2,1,2}^H$	137.8	$b_{4,3,2}^{C,\text{out}}$	1857.24
$t_{2,2,1}^H$	160.34047781513647	$b_{5,1,2}^{C,\text{out}}$	367.27999999999986
$t_{3,5,1}^H$	85.5953615126148	$b_{5,4,2}^{C,\text{out}}$	3056.91
$t_{4,3,2}^H$	148.9		
$t_{5,1,2}^H$	91.18281635356576		
$t_{5,4,2}^H$	100.01169124992158		
$t_{2,1,2}^C$	129.79605263157896		
$t_{2,2,1}^C$	221.7		
$t_{3,5,1}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,1,2}^C$	90.20394736842104		
$t_{5,4,2}^C$	176.7		
$b_{2,1,2}^{H,\text{in}}$	1991.13		
$b_{2,2,1}^{H,\text{in}}$	1449.960117011701		
$b_{3,5,1}^{H,\text{in}}$	2173.1256548727506		
$b_{4,3,2}^{H,\text{in}}$	3415.8599999999997		
$b_{5,1,2}^{H,\text{in}}$	342.51684171042746		
$b_{5,4,2}^{H,\text{in}}$	3178.013158289574		
$b_{2,1,2}^{H,\text{out}}$	1460.68		
$b_{2,2,1}^{H,\text{out}}$	802.750117011701		
$b_{3,5,1}^{H,\text{out}}$	628.83565487275		
$b_{4,3,2}^{H,\text{out}}$	1876.1399999999996		
$b_{5,1,2}^{H,\text{out}}$	112.9668417104275		
$b_{5,4,2}^{H,\text{out}}$	1543.1631582895745		
$b_{2,1,2}^{C,\text{in}}$	318.27000000000004		
$b_{2,2,1}^{C,\text{in}}$	705.16		
$b_{3,5,1}^{C,\text{in}}$	1296.87		
$b_{4,3,2}^{C,\text{in}}$	317.52		
$b_{5,1,2}^{C,\text{in}}$	137.72999999999993		
$b_{5,4,2}^{C,\text{in}}$	1422.0600000000002		

3 *RecLMTD* Tangent Points

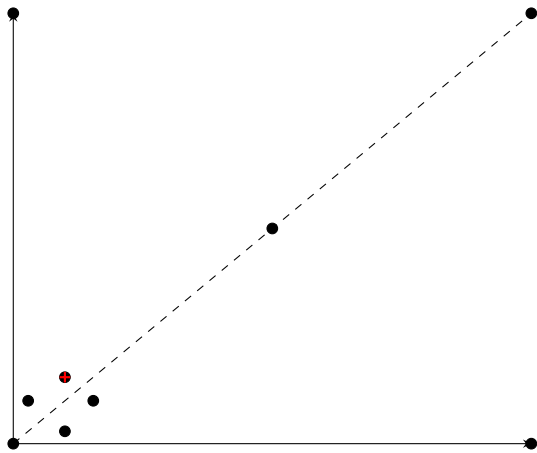
- (2, 1, 2)
 - Value: 0.023339
 - Error: 0.006919464988083
 - Relative error: 0.228675039265966
- (2, 2, 1)
 - Value: 0.021686
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000054
- (3, 5, 1)
 - Value: 0.039171
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000001
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000874453881
 - Relative error: 0.000066919610990
- (5, 1, 2)
 - Value: 0.020327
 - Error: 0.007322227802647
 - Relative error: 0.264829009596219
- (5, 4, 2)
 - Value: 0.055257
 - Error: 0.006534283138936
 - Relative error: 0.105747395683138
- $cu, 1$
 - Value: 0.010901
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 3$
 - Value: 0.016938
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 5$
 - Value: 0.021054
 - Error: 0.000320342867718
 - Relative error: 0.014986994547027



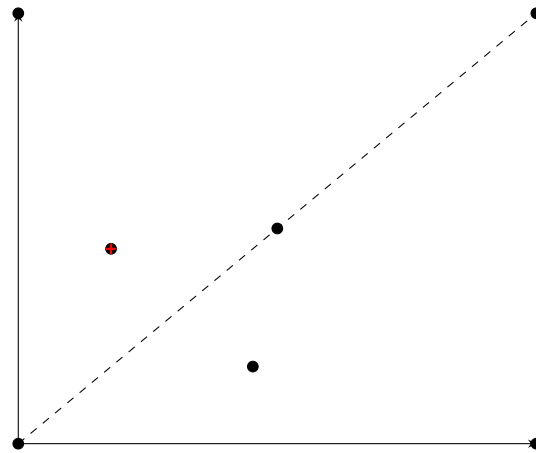
(a) *RecLMTD* - (2, 1, 2)



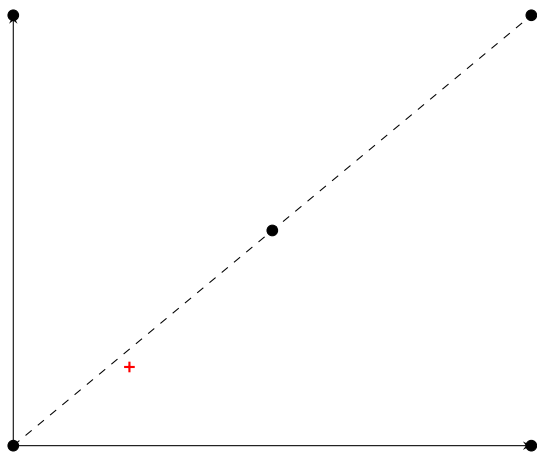
(b) *RecLMTD* - (2, 2, 1)



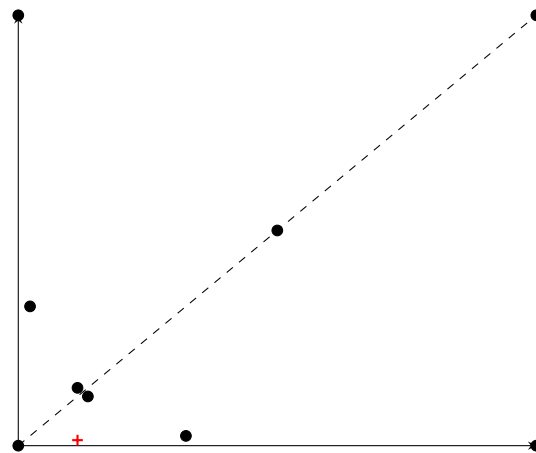
(c) *RecLMTD* - (3, 5, 1)



(d) *RecLMTD* - (4, 3, 2)



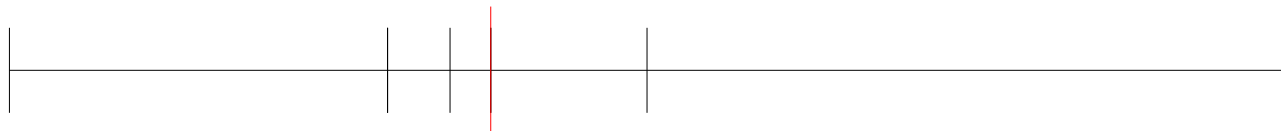
(e) *RecLMTD* - (5, 1, 2)



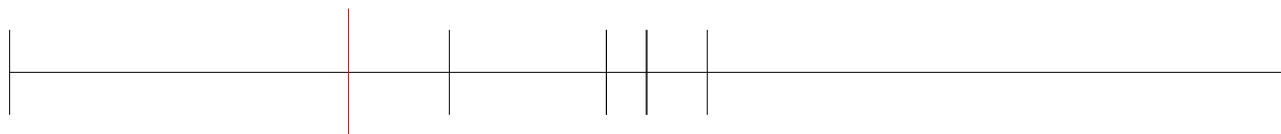
(f) *RecLMTD* - (5, 4, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - $cu, 3$



(c) *RecLMTD* - $cu, 5$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (2, 1, 2)
 - Value: 1991.13
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.960117011701
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2173.1256548727506
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 1, 2)
 - Value: 342.51684171042746
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 3178.013158289574
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.430917142996387e-16

4.2 $b^{H,\text{out}}$

- (2, 1, 2)
 - Value: 1460.68
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 802.750117011701
 - Absolute error: 131.30893471408172
 - Relative error: -0.1405788365001904
- (3, 5, 1)

- Value: 628.83565487275
 - Absolute error: 191.67372334150184
 - Relative error: -0.23360333035882974
- (4, 3, 2)
 - Value: 1876.1399999999996
 - Absolute error: 4.547473508864641e-13
 - Relative error: -2.4238455066597593e-16
- (5, 1, 2)
 - Value: 112.9668417104275
 - Absolute error: 44.05502996219823
 - Relative error: -0.2805662007013162
- (5, 4, 2)
 - Value: 1543.1631582895745
 - Absolute error: 54.81809246479952
 - Relative error: -0.03430459051939504

4.3 $b^{C,in}$

- (2, 1, 2)
 - Value: 318.27000000000004
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 1, 2)
 - Value: 137.72999999999993
 - Absolute error: 0.0
 - Relative error: 0.0

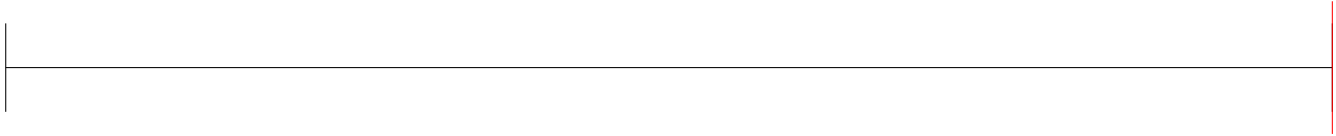
- (5, 4, 2)
 - Value: 1422.0600000000002
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (2, 1, 2)
 - Value: 848.72
 - Absolute error: 160.2168388157893
 - Relative error: 0.2327031273759437
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2841.1600000000003
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.24
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 1, 2)
 - Value: 367.27999999999986
 - Absolute error: 160.21683881578946
 - Relative error: 0.773758296258479
- (5, 4, 2)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (2, 1)$



(a) $t^{(H)} - (3, 1)$



(a) $t^{(H)} - (4, 2)$



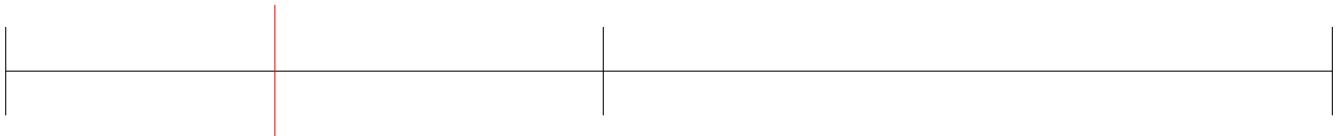
(a) $t^{(H)} - (5, 2)$



(a) $t^{(H)} - (5, 2)$



(a) $t^{(H)} - (2, 1, 2)$



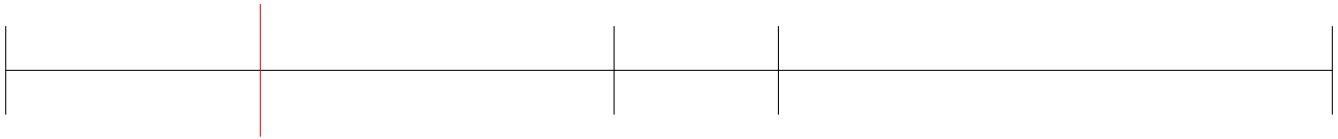
(a) $t^{(H)} - (2, 2, 1)$



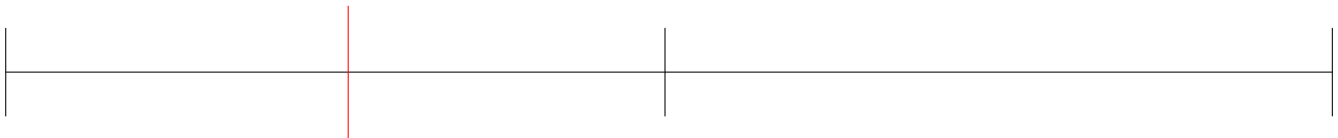
(a) $t^{(H)} - (3, 5, 1)$



(a) $t^{(H)} - (4, 3, 2)$



(a) $t^{(H)} - (5, 1, 2)$



(a) $t^{(H)} - (5, 4, 2)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (2, 2)$



(a) $t^{(C)} - (5, 2)$



(a) $t^{(C)} - (3, 3)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (4, 3)$



(a) $t^{(C)} - (2, 1, 2)$



(a) $t^{(C)} - (2, 2, 1)$



(a) $t^{(C)} - (3, 5, 1)$



(a) $t^{(C)} - (4, 3, 2)$



(a) $t^{(C)} - (5, 1, 2)$



(a) $t^{(C)} - (5, 4, 2)$

5 Area Breakpoints

5.1 A

- $(2, 1, 2)$
 - Value: 3.303646
 - Error: 11.261560717287871
 - Relative error: 0.773182320759116
- $(2, 2, 1)$
 - Value: 16.512573
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(3, 5, 1)$
 - Value: 71.165974
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(4, 3, 2)$
 - Value: 23.668856
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(5, 1, 2)$
 - Value: 1.944268
 - Error: 3.545126131875561
 - Relative error: 0.645813737267882
- $(5, 4, 2)$
 - Value: 106.279002
 - Error: 0.000000000000128
 - Relative error: 0.000000000000001

5.2 A_{CU}

- $cu, 1$
 - Value: 7.527593
 - Error: 0.000000000000041
 - Relative error: 0.000000000000005
- $cu, 3$
 - Value: 16.738997
 - Error: 0.000000000000011
 - Relative error: 0.000000000000001
- $cu, 5$

- Value: 6.717166
- Error: 5.544162443527119
- Relative error: 0.452166543807991



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (3, 5, 1)$



(a) $A^\beta - (4, 3, 2)$



(a) $A^\beta - (5, 1, 2)$



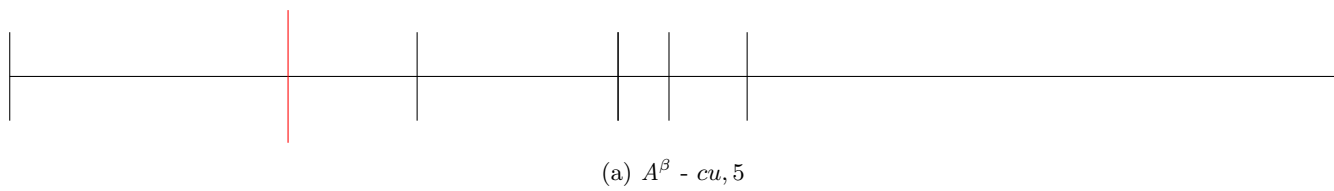
(a) $A^\beta - (5, 4, 2)$



(a) $A^\beta - cu, 1$



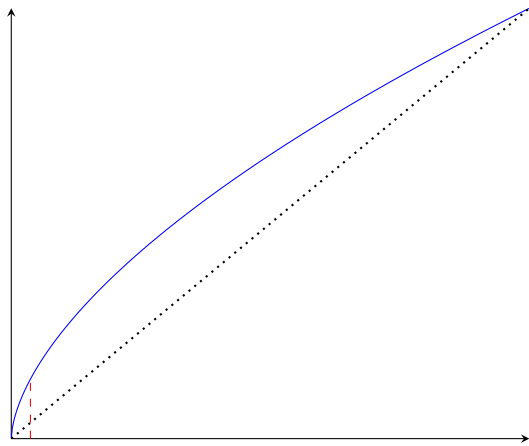
(a) $A^\beta - cu, 3$



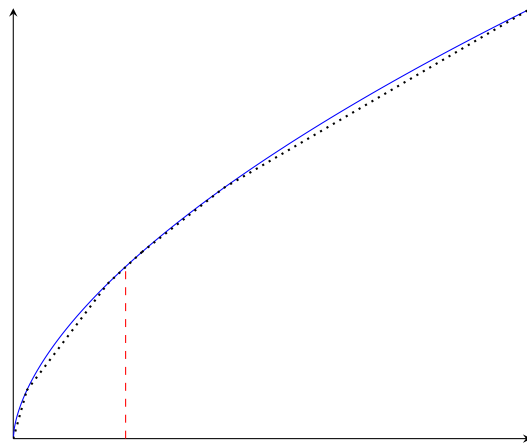
6 A^β Breakpoints

- (2, 1, 2)
 - q : 3.303646
 - $\hat{q}^\beta$: 0.547565
 - q^β : 2.048312
 - Breakpoints: [0, 89.41176470588236]
 - Error: 1.500747125526037
 - Relative error: 0.732675008665604
- (2, 2, 1)
 - q : 16.512573
 - $\hat{q}^\beta$: 5.363812
 - q^β : 5.378843
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 18.890814020083003, 30.93278282491032, 76.14235294117645]
 - Error: 0.015030567552889
 - Relative error: 0.002794386843874
- (3, 5, 1)
 - q : 71.165974
 - $\hat{q}^\beta$: 12.265237
 - q^β : 12.923003
 - Breakpoints: [0, 30.750913778350093, 44.57376487924321, 181.68117647058827]
 - Error: 0.657765995036538
 - Relative error: 0.050898851087590
- (4, 3, 2)
 - q : 23.668856
 - $\hat{q}^\beta$: 6.675847
 - q^β : 6.675847
 - Breakpoints: [0, 19.92063386984238, 23.668855770059224, 181.14352941176475]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (5, 1, 2)
 - q : 1.944268
 - $\hat{q}^\beta$: 0.322254
 - q^β : 1.490231
 - Breakpoints: [0, 89.41176470588236]
 - Error: 1.167977324146728
 - Relative error: 0.783755742349608
- (5, 4, 2)

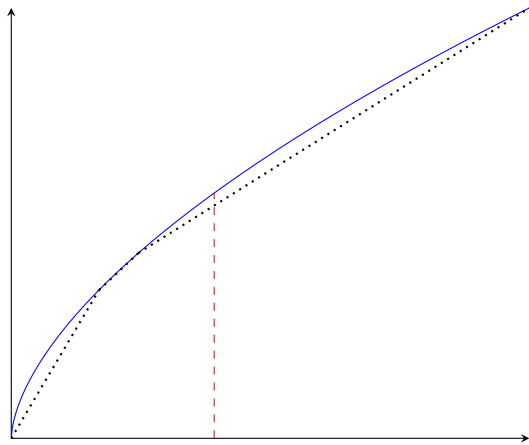
- q : 106.279002
- $\hat{q}^\beta$: 15.629519
- q^β : 16.438738
- Breakpoints: [0, 11.133625686778567, 51.43476571725509, 192.3352941176471]
- Error: 0.809218906488276
- Relative error: 0.049226339224178
- $cu, 1$
 - q : 7.527593
 - $\hat{q}^\beta$: 3.357327
 - q^β : 3.357327
 - Breakpoints: [0, 7.5275930538960525, 22.75735506822926]
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000000
- $cu, 3$
 - q : 16.738997
 - $\hat{q}^\beta$: 5.422976
 - q^β : 5.422976
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 14.376341772690381, 16.738997089244645, 128.44338870924773]
 - Error: 0.0000000000000001
 - Relative error: -0.0000000000000000
- $cu, 5$
 - q : 6.717166
 - $\hat{q}^\beta$: 2.692325
 - q^β : 3.135535
 - Breakpoints: [0, 9.832103793393394, 14.682070969056184, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.443209890727134
 - Relative error: 0.141350646838335



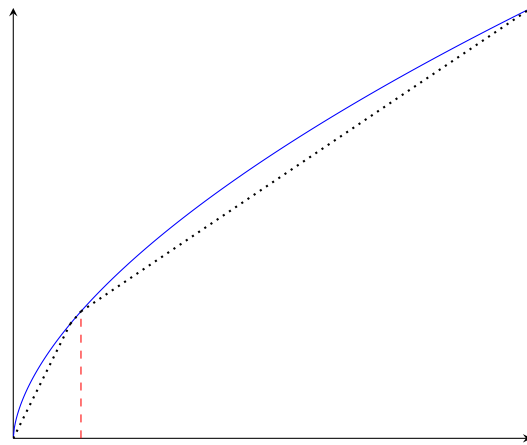
(a) $q^\beta - (2, 1, 2)$



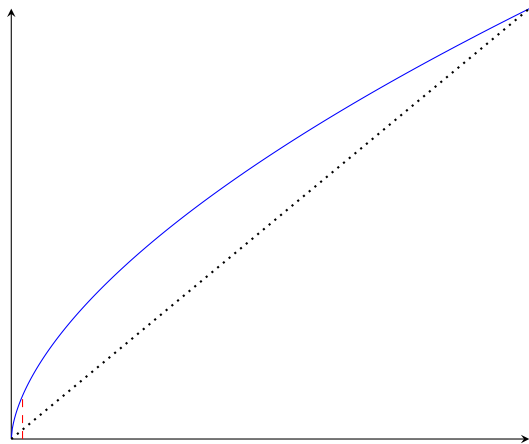
(b) $q^\beta - (2, 2, 1)$



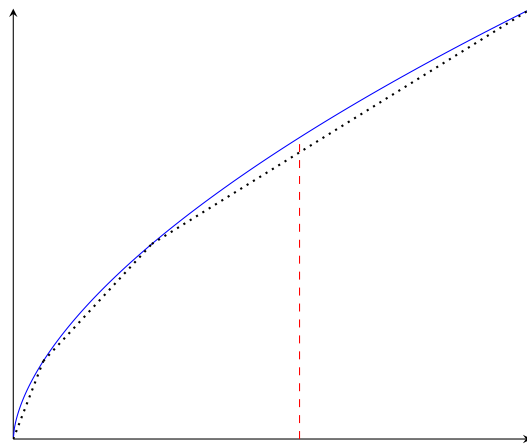
(c) $q^\beta - (3, 5, 1)$



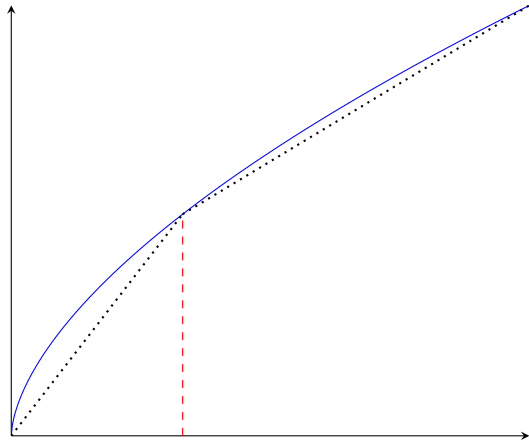
(d) $q^\beta - (4, 3, 2)$



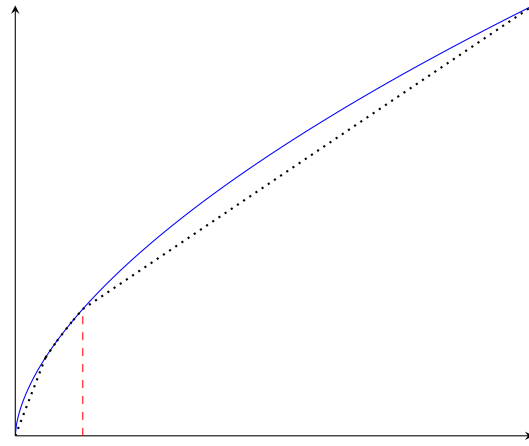
(e) $q^\beta - (5, 1, 2)$



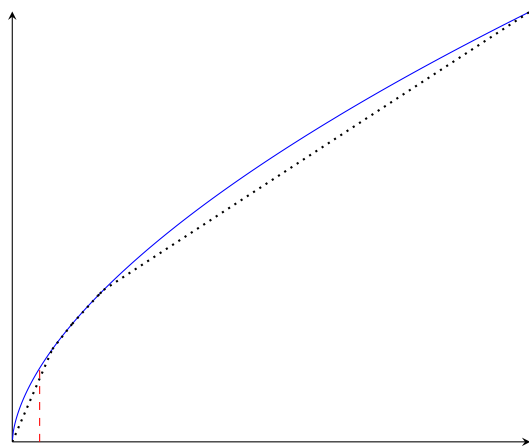
(f) $q^\beta - (5, 4, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 3$



(c) $q^\beta - cu, 5$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,4,2}$	0.0	$A_{2,4,2}^\beta$	0.0
$z_{1,1,2}$	-0.0	$z_{4,5,1}$	-0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{4,5,2}$	-0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,1,1}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,2}$	0.0	$z_{5,2,2}$	-0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,3,2}$	-0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,1}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,1}$	-0.0	$z_{5,5,2}$	-0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,2,2}$	-0.0	$z_{cu,2}$	-0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,3,1}$	-0.0	$z_{cu,4}$	-0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,4,1}$	0.0	$z_{hu,2}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,2}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	0.0	$z_{hu,4}$	-0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,2}$	0.0	$z_{hu,5}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$A_{1,1,2}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{3,5,2}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{2,1,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,1}$	-0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,3,1}$	0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,2,2}$	0.0	$dt_{1,5,1}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,3,1}$	0.0	$dt_{1,5,2}$	10.0
$A_{hu,5}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{1,1,1}$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,1}$	111.1
$A_{1,1,2}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,2,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,2}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,2}$	10.0
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,5,1}$	0.0	$A_{5,4,1}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{2,5,2}$	155.6
$A_{2,1,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{2,5,3}$	111.1
$A_{2,2,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,3,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,3,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,4,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,4,2}$	0.0	$A_{hu,3}$	0.0	$dt_{3,2,2}$	111.1
$A_{2,5,1}$	0.0	$A_{hu,4}$	0.0	$dt_{3,2,3}$	111.1
$A_{2,5,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,1}$	161.1
$A_{3,1,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,3,2}$	10.0
$A_{3,1,2}$	0.0	$dt_{1,1,2}$	10.0	$dt_{3,3,3}$	188.9
$A_{3,2,1}$	0.0	$dt_{1,1,3}$	33.3	$dt_{3,4,1}$	144.5
$A_{3,2,2}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,4,2}$	10.0
$A_{3,3,1}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{3,4,3}$	144.5
$A_{3,3,2}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{3,5,3}$	133.4
$A_{3,4,1}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,1}$	122.2
$A_{3,4,2}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,1,2}$	10.0
$A_{3,5,2}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,1,3}$	100.0
$A_{4,1,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,1}$	122.2
$A_{4,1,2}$	0.0	$dt_{1,4,2}$	10.0	$dt_{4,2,2}$	10.0
$A_{4,2,1}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,2,3}$	106.1

Variable	Value	Variable	Value	Variable	Value
$dt_{4,3,1}$	122.2	$f_{1,5,1}^H$	0.0	$f_{5,4,1}^H$	0.0
$dt_{4,4,1}$	122.2	$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	17.7
$dt_{4,4,2}$	122.2	$f_{2,1,1}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,4,3}$	94.5	$f_{2,2,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,1}$	122.2	$f_{2,3,1}^H$	0.0	$f_{1,1,2}^C$	0.0
$dt_{4,5,2}$	10.0	$f_{2,3,2}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,4,1}^H$	4.774527452745275	$f_{1,2,2}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,4,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,1}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,2}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,3}$	74.66004781348391	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,1,2}^H$	14.8	$f_{1,5,1}^C$	0.0
$dt_{5,3,2}$	53.241998311284874	$f_{3,2,1}^H$	5.214090626939786	$f_{1,5,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,2,2}^H$	0.0	$f_{2,1,1}^C$	0.0
$dt_{5,4,1}$	116.7	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,5,1}$	105.6	$f_{3,3,2}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,5,2}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,4,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{cu,2}$	97.8	$f_{3,5,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,1}^H$	12.6	$f_{2,5,1}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,1,2}^H$	0.0	$f_{2,5,2}^C$	13.9
$dt_{hu,2}$	18.3	$f_{4,2,1}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,2,2}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,3,1}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$f_{1,1,1}^H$	0.0	$f_{4,4,2}^H$	0.0	$f_{3,3,1}^C$	0.0
$f_{1,1,2}^H$	0.0	$f_{4,5,1}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,2,1}^H$	8.8	$f_{4,5,2}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,3,2}^H$	8.8	$f_{5,2,2}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,1,2}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,2,1}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,2,2}^C$	0.0	$\text{RecLMTD}_{3,1,2}$	0.1	$q_{1,1,2}$	0.0
$f_{4,3,1}^C$	8.4	$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,2,1}$	0.0
$f_{4,4,1}^C$	17.3	$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,2}$	0.0
$f_{4,4,2}^C$	0.0	$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,3,1}$	0.0
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,2}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.1	$q_{1,4,1}$	0.0
$f_{5,1,1}^C$	7.6	$\text{RecLMTD}_{3,4,2}$	0.1	$q_{1,4,2}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{3,5,2}$	0.1	$q_{1,5,1}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.020350921041481694	$q_{1,5,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$q_{2,1,1}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.1	$q_{2,2,2}$	0.0
$f_{5,4,1}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$q_{2,3,1}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.013066354534947922	$q_{2,3,2}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.02230651890829312	$q_{2,4,1}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,4,2}$	0.024030191698075007	$q_{2,4,2}$	0.0
$\text{RecLMTD}_{1,1,2}$	0.1	$\text{RecLMTD}_{4,5,1}$	0.1	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02554670259920889	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.01228852169748186	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.03070641897615819	$\text{RecLMTD}_{5,2,2}$	0.1	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.1	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028822002049121505	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,1}$	0.1	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$q_{3,4,1}$	0.0
$\text{RecLMTD}_{2,1,1}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.009469696969696968	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115769086	$q_{3,5,2}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.1	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.025165980106347276	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.04280840229803139	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,3,1}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.03118293144804949	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$q_{1,1,1}$	0.0	$q_{4,4,2}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{4,5,1}$	0.0	$t_{2,5,1}^H$	198.8575471698107	$t_{1,3,2}^C$	203.68591798578967
$q_{4,5,2}$	0.0	$t_{2,5,2}^H$	184.96686427216463	$t_{1,4,1}^C$	176.7
$q_{5,1,1}$	0.0	$t_{3,1,1}^H$	123.99729729729722	$t_{1,4,2}^C$	102.85953757225434
$q_{5,2,1}$	0.0	$t_{3,1,2}^H$	122.35608108108093	$t_{1,5,1}^C$	148.97551619911107
$q_{5,2,2}$	0.0	$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3
$q_{5,3,1}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,1,1}^C$	129.7960526315789
$q_{5,3,2}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,4,1}$	0.0	$t_{3,3,2}^H$	65.6	$t_{2,3,1}^C$	221.1
$q_{5,5,1}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,2}$	0.0	$t_{3,4,2}^H$	65.6	$t_{2,4,1}^C$	85.63275904837246
$q_{cu,2}$	0.0	$t_{3,5,2}^H$	122.35608108108096	$t_{2,4,2}^C$	112.86184971098261
$q_{cu,4}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,1}^C$	139.21155520203027
$q_{hu,1}$	0.0	$t_{4,1,2}^H$	148.9	$t_{2,5,2}^C$	93.3
$q_{hu,2}$	0.0	$t_{4,2,1}^H$	271.1	$t_{3,1,1}^C$	160.0
$q_{hu,3}$	0.0	$t_{4,2,2}^H$	148.9	$t_{3,1,2}^C$	60.0
$q_{hu,4}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,1}^C$	115.6
$q_{hu,5}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$t_{1,1,1}^H$	160.0	$t_{4,4,2}^H$	148.9	$t_{3,3,1}^C$	221.1
$t_{1,1,2}^H$	133.9147727272728	$t_{4,5,1}^H$	271.1	$t_{3,3,2}^C$	37.8
$t_{1,2,1}^H$	160.0	$t_{4,5,2}^H$	148.9	$t_{3,4,1}^C$	176.7
$t_{1,2,2}^H$	160.0	$t_{5,1,1}^H$	179.34745762711862	$t_{3,4,2}^C$	82.2
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{3,5,2}^C$	93.3
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	121.42829734414263	$t_{4,1,1}^C$	160.0
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,1,2}^C$	60.0
$t_{1,4,2}^H$	160.0	$t_{5,3,2}^H$	65.60116645009666	$t_{4,2,1}^C$	221.7
$t_{1,5,1}^H$	160.0	$t_{5,4,1}^H$	126.72824858757063	$t_{4,2,2}^C$	115.6
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,3,1}^C$	221.1
$t_{2,1,1}^H$	198.85754716981148	$t_{5,5,2}^H$	126.72824858757063	$t_{4,4,1}^C$	176.7
$t_{2,2,2}^H$	187.8424528301887	$t_{1,1,1}^C$	160.0	$t_{4,4,2}^C$	82.2
$t_{2,3,1}^H$	159.0169811320755	$t_{1,1,2}^C$	144.10854085872566	$t_{4,5,1}^C$	204.4
$t_{2,3,2}^H$	184.96686427216463	$t_{1,2,1}^C$	115.6	$t_{4,5,2}^C$	93.3
$t_{2,4,1}^H$	248.9	$t_{1,2,2}^C$	115.6	$t_{5,1,1}^C$	160.0
$t_{2,4,2}^H$	184.96686427216463	$t_{1,3,1}^C$	221.1	$t_{5,2,1}^C$	115.6

Variable	Value	Variable	Value	Variable	Value
$t_{5,2,2}^C$	115.6	$b_{4,1,1}^{H,in}$	3415.86	$b_{2,5,1}^{H,out}$	0.0
$t_{5,3,1}^C$	221.1	$b_{4,1,2}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	0.0
$t_{5,3,2}^C$	37.8	$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0
$t_{5,4,1}^C$	176.7	$b_{4,2,2}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	1810.8699999999994
$t_{5,5,1}^C$	204.4	$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	1182.0343451272495
$t_{5,5,2}^C$	124.01079136690639	$b_{4,4,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	0.0	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0
$b_{1,1,2}^{H,in}$	0.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	1408.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	1408.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,1}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	3520.5299999999997	$b_{4,3,1}^{H,out}$	0.0
$b_{2,1,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,1,2}^{H,out}$	0.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	1408.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	1188.379882988299	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	1408.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	1810.8699999999994	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,1}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	1182.0343451272495	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	3520.53
$b_{3,2,2}^{H,in}$	0.0	$b_{2,1,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,1,2}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	1188.379882988299	$b_{1,2,2}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{1,3,2}^{C,in}$	0.0	$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0
$b_{1,4,1}^{C,in}$	0.0	$b_{5,3,1}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0
$b_{1,4,2}^{C,in}$	0.0	$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0
$b_{1,5,1}^{C,in}$	0.0	$b_{5,4,1}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0
$b_{1,5,2}^{C,in}$	0.0	$b_{5,5,1}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	1857.24
$b_{2,1,1}^{C,in}$	0.0	$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	3056.91
$b_{2,2,2}^{C,in}$	705.16	$b_{1,1,1}^{C,out}$	0.0	$b_{4,4,2}^{C,out}$	0.0
$b_{2,3,1}^{C,in}$	0.0	$b_{1,1,2}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	1216.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	1296.87	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,1}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	705.16		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	1296.87		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,3,1}^{C,in}$	1857.24	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	3056.91	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	1216.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		